Software Test Plan (STP) - Project Management System

Project: Project Management System

Version: 1.0

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1. Introduction

Purpose: This document defines the test plan for the Project Management System (PMS) v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers PMS features such as authentication and role-based access, project management, task/issue management, the Kanban board, workflow management, estimation, sprint/milestone management, the dashboard, notifications, comments/collaboration, search/filter, reporting, and the activity log. Multi-organization administration, native mobile applications, and deep third-party integrations (e.g., live Slack/GitHub webhooks) are excluded, consistent with the V1.0 scope.

References: PMS Project Proposal v1.0; PMS SRS v1.0 (IEEE 830); PMS Requirements document (Team 11).

Definitions: PMS (Project Management System), RBAC (Role-Based Access Control), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), FR (Functional Requirement), NFR (Non-Functional Requirement).

2. Test Items

- Authentication & Role-Based Access Control module
- Project Management module
- Task/Issue Management module
- Kanban Board module
- Workflow Management module
- Estimation module
- Sprint/Milestone Management module
- Visibility & Dashboard module
- Notifications module
- Comments & Collaboration module
- Search & Filter module
- Reporting module
- Activity Log module

3. Features to be Tested

Features mapped to SRS requirement IDs:

- FR1: User registration, login/logout, password reset, role-based access (Admin, Project Manager, Team Member)
- FR2: Create, edit, archive, delete projects; add/remove team members
- FR3: Create, edit, delete, and assign tasks (title, description, priority, due date, labels)

- FR4: Kanban board rendering and drag-and-drop status updates, including rejection of invalid moves
- FR5: Custom workflow state and transition definition per project
- FR6: Story-point/time estimation and estimated-vs-actual effort tracking
- FR7: Sprint/milestone creation, backlog management, and sprint closure with carry-over
- FR8: Dashboard, burndown/velocity charts, task distribution by status and assignee
- FR9: Notifications on assignment, status change, comments, and approaching deadlines
- FR10: Comments and file/document attachments on tasks
- FR11: Keyword search and multi-criteria filtering (assignee, priority, status, label, sprint)
- FR12: Progress, velocity, and workload report generation
- FR13: Activity log capture and chronological display per task/project
- NFR1: Page load and Kanban update response time ≤ 2 seconds under normal load
- NFR3: Password hashing, RBAC enforcement, HTTPS, and protection against SQL injection, XSS, and CSRF
- NFR5: 99% uptime and graceful error handling

4. Features Not to be Tested

- Multi-organization/workspace administration (out of V1.0 scope)
- Native mobile applications (web UI is responsive for tablet only)
- Live third-party integrations such as Slack/GitHub webhooks, beyond the basic REST API
- Underlying database engine internals (PostgreSQL/MySQL/MongoDB) and hosting infrastructure

5. Test Approach / Strategy

Levels:

- Unit tests (module-level)
- Integration tests (module ↔ module, e.g., Kanban ↔ Workflow, Sprint ↔ Estimation)
- System tests (end-to-end PMS functionality)
- Acceptance tests (evaluator/course walkthrough)

Types:

- Functional testing (core features per FR1–FR13)
- Regression testing (workflow transition rules, RBAC enforcement across releases)
- Performance testing (response time, load per NFR1)
- Usability testing (drag-and-drop clarity, responsive layout per NFR4)

Entry Criteria: Stable build delivered, test data available, test environment ready

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied

5.1 Security Validation

- Validate role-based access control for Admin, Project Manager, and Team Member roles
- Verify password hashing (bcrypt) and no plaintext password storage or logging
- HTTPS verification for all client-server communication
- Input validation and fuzzing for task/project fields (title, description, comments)
- Checks against SQL injection, XSS, and CSRF on all forms and API endpoints

6. Test Environment

Software: PMS application (Python/Django or MERN stack); supported browsers — Chrome, Firefox, Edge, Safari; relational database (PostgreSQL/MySQL) or MongoDB

Tools: Selenium (UI automation), Postman (API testing), JMeter (performance testing), GitHub Issues/Jira (defect tracking)

Test Data: Dummy user accounts for each role, sample projects, tasks, workflows, and sprints

7. Test Schedule

Milestones (aligned to the project plan of work):

- Test case design: Week 7 (alongside estimation & sprint module implementation)
- Environment setup: Week 8
- Test execution start: Week 9 (integration testing across modules)
- Test execution end: Week 10
- Acceptance testing / demo preparation: Week 10

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Shivansh Sharma	Prepare test plan, coordinate execution; tests Authentication/RBAC, Project Management, Activity Log
Test Engineer	Sheetal S Bevinakatti	Design & execute test cases for Task/Issue Management, Kanban Board, Workflow Management
Developer	Shibi R Shetty	Design & execute test cases for Estimation, Sprint Management, Notifications, Comments & Collaboration
Product Owner	Shivanagouda I Patil	Design & execute test cases for Dashboard, Search & Filter, Reporting

10. Risks and Mitigation

Risk	Mitigation
Delay in finalizing tech stack (Django vs. MERN)	Finalize stack by end of Week 3 as per the plan of work
Incomplete workflow/Kanban integration by test time	Prioritize FR4/FR5 implementation in Weeks 6, track progress against the weekly plan
Team member unavailability	Shared documentation and cross-familiarity with owned modules
Dependency on optional third-party APIs (Slack/GitHub)	Treat as out of scope/stubbed for V1.0; do not block core testing

11. Assumptions & Dependencies

- Users have access to a modern web browser and a stable internet connection
- Third-party integrations (Slack, GitHub, email) depend on the availability and API stability of those external services
- A small-to-moderate concurrent user base is assumed, consistent with a mini-project deployment

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment unavailable for more than 4 hours
- Build is too unstable (blocks more than 30% of test cases)

Resume testing if:

- Blocking defects are resolved
- Environment is stabilized

13. Test Case Management & Traceability

The RTM (SRS Appendix C) ensures mapping of SRS requirements to test cases. Example:

- FR1 (User Authentication) → TC-Auth-01, TC-Auth-02
- FR3 (Task/Issue Management) → TC-Task-01, TC-Task-02
- FR4 (Kanban Board) → TC-Kanban-01, TC-Kanban-02
- FR6 (Estimation) → TC-Est-01
- FR8 (Visibility & Dashboard) → TC-Dash-01

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Weekly execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Shivansh Sharma	Shivansh Sharma /17.09.2026
Dev Lead	Shibi R Shetty	Shibi R Shetty/17.09.2026
Product Owner	Shivanagouda I Patil	Shivanagouda I Patil /17.09.2026